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Invited paper

Experimental research on damage identification of fully free beams based on modal shapes

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ABSTRACT

Ship damage identification is a new field of ship research, which is of great significance to damage control, life and property safety and strategic decision-making, but there is no effective method for ship damage identification. The structural system changes caused by ship damage are usually represented by the change of system mass, stiffness and energy dissipation characteristics, and then the change of structural dynamic characteristic parameters, which provides a theoretical basis for ship damage identification. In this paper, the ship is simplified as a beam model with fully free boundary conditions. Based on the dynamic characteristic parameter of the modal shape, the damage identification of the ship under six damage conditions is carried out through numerical simulation and experimental verification by using COMAC and the derived indexes of curvature mode and modal flexibility. The results show that the first four orders of curvature mode difference mean index and the curvature index of modal flexibility change rate can accurately locate the damage of hull beam structure; The damage degree of hull beam can be identified by the change of curvature mode difference index, but there are some errors in this method; The curvature index of modal flexibility change rate is positively correlated with the damage degree, which can reflect the damage degree of the structure. The research results of this paper can provide new methods and ideas for hull beam damage identification.

1. Introduction

Ship damage will bring about local strength and total strength reduction to the ship, and once the damage is serious, it will affect the structural safety of the ship and its subsequent use. For civilian vessels, collisions and groundings are the main sources of damage. Recently, in the Gulf of Catalonia, a cargo ship named OS35 collided with the LNG tanker AdamLNG, causing a nearly 10-meter-long crack in the hull of the OS35 cargo ship. Failure to detect the damage in time led to more serious damage to the vessel, which eventually went aground in the water. However, the collision not only brought about the problem of damage to the vessel, but also led to the problem of oil spills, which had a great impact on the marine ecosystem. For military ships, explosions and weapons attacks are the main source of damage. Non-contact underwater explosions, such as bottom mines and torpedoes, can cause rapid and violent deformation of the hull, causing the ship's main beam to bend and collapse, which was the most common cause of destroyer losses during World War II. Once the ship is attacked and damaged by water, it is necessary to judge the extent of damage in time whether it is still damaging the management value and whether it is necessary to abandon the ship. Through the above cases, no matter the civilian ship or the military ship, the hull structural damage changes will cause the overall damage of the ship and the significant loss of life and property. Therefore, an effective method to identify ship structural damage is urgently needed in engineering applications.

At present, most studies on ship damage focus on the forward process of external load forming damage (Zhang et al., 2012; Martin et al., 2012; McKown et al., 2008), that is, the load input of ship structure is known and the structural system is used to solve the overall or local response of ship (Guo et al., 2019; Jun et al., 2019), but there is a lack of technical means for real-time identification of structural damage, that is, the structural changes of the system can be deduced by the known input and response. This damage identification technology has been widely used in civil engineering, aeronautical engineering and bridge engineering. Vibration-based health monitoring of civil structures relies on the repeated identification of the dynamic characteristics of structures from the vibration data that is only output. Because the dynamic strain levels encountered under environmental excitation are very small, it is

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challenging to accurately identify strain mode modes in dense grids. Anastasopoulos et al. (2018) proposed a solution to this challenge that allows earlier identification of the presence and location of damage. Kahya et al. (2022, 2021a, 2021b, 2021c) analyzed the free vibration of a tapered beam with damage and searched for the damage location by modal curvature variation. Then, the damage detection of composite laminated beams based on vibration measurement, automatic model updating, shear deformation finite element and harmonious search optimization were studied respectively. Altunisik et al. (2019) obtained the inherent frequencies and corresponding vibration patterns of steel cantilever beams with circular-hollow section and box section for six damage conditions by means of analytical and experimental models. On this basis, the modal curvature method and modal flexibility method were used to localize the cracks in the beams. Simsek et al. (2022) proposed an effective method to detect damage in arbitrary lay-up composite laminated beams, which better identifies single and multiple damage in anisotropic laminated composite beams and outperforms other algorithms in terms of accuracy and computational cost. Genc et al. (2021) investigated the experimental and numerical evaluation of the intrinsic frequencies and corresponding modal vibrations of cross-laminated timber (CLT) beams under a range of damage scenarios. Cao and Qiao (2009) proposed a new Laplace format and based on it, an improved damage identification algorithm is proposed. Compared with the standard Laplace operator, this algorithm has the advantages of high noise insensitivity, clear damage display and visualization of damage presentation, which is a promising damage identification algorithm. Ciambella et al. (2019) considered weakly damaged continuous beams and established an inverse problem for damage localization based on modal curvature alone, demonstrating that modal curvature can be used for accurate damage localization provided that proper filtering is performed. As a beam structure, the dynamic characteristics of ship are similar to those of other beam structures. The above literatures have reference significance for the research of ship structural damage identification.

Unlike most of the current studies on ship damage forward, this paper solves the inverse problem for ship damage identification by using COMAC index and the derived indexes of curvature mode and modal flexibility for hull beam damage identification. Since the total vibration modes of the beam with variable section are similar to those of the beam with uniform cross-section, the hull beam with variable section is simplified to be a uniform beam for the convenience of model processing, so as to verify the effectiveness of the damage identification principle and method of hull beam under the condition of full free boundary. In this paper, different damage conditions are set, and the modal data of hull beams are obtained by finite element modal analysis. The damage identification of hull beams is carried out by using the indexes such as COMAC, curvature mode difference, curvature mode ratio and modal flexibility change rate, and the stability and accuracy of each index for ship damage identification are analyzed. Through model tests, the modal parameters of intact and damaged free beams were measured by single excitation method, and damage location and quantitative identification were completed based on the proxy model. This paper verifies that the above indexes are valid for hull beam damage identification from the modal vibration pattern of hull beam structure, and provides a new method and idea for hull beam damage identification.

2. Damage identification theory of dynamic characteristic parameters

When the ship structure is damaged, its stiffness and mass will also change, which will cause the change of dynamic characteristic parameters, such as the natural frequency of the structure and the modal shape. The test accuracy of the mode is lower than that of the natural frequency, but the mode contains more damage information. In this paper, the modal shape method will be used to identify the damage of hull beams. The damage recognition theory of hull beams based on the modal shape is given below.

2.1. Damage identification theory based on COMAC

The modal assurance criterion (MAC) uses the orthogonality of the vibration modes before and after damage to identify the damage. On the basis of this, it is improved to use the multi-order vibration modes to determine the damage and establish the connection with the position coordinate, which is the coordinate modal assurance criterion (COMAC). COMAC index definition is expressed as follows (Lieven, 1988):

$$COMAC = \frac{\left(\sum_{r=1}^{n} |\varphi_r^u(i)\varphi_r^d(i)|\right)^2}{\sum_{r=1}^{n} \varphi_r^{u2}(i) \sum_{r=1}^{n} \varphi_r^{d2}(i)} \tag{1}$$

Where: n is the total number of modes of hull beams used to identify damage;

$\varphi_r^u(i)$ is the mode vector of the r-order mode at the i-position of the undamaged hull beam structure;

$\varphi_r^d(i)$ is the mode vector of the r-order mode at the i-position of the damaged hull beam structure.

The value of COMAC ranges from 0 to 1,when$\varphi_r^u(i)$=$\varphi_r^d(i)$, COMAC=1, indicating that the structure is not damaged. When the COMAC value is low, it indicates that the structure is damaged at position i. According to the expression of the index, when the selected mode number is 1, the COMAC value is always 1, so the selected mode number is at least 2. In this paper, the first two, first three and first four modes of hull beams are calculated and compared respectively to analyze the sensitivity and accuracy of the modes to damage identification based on COMAC method. In practice, this value can be used flexibly to locate and identify damage units more clearly.

2.2. Damage identification theory based on curvature mode

According to the mechanics of materials, the basic formula of one-dimensional beam bending deformation is as follows:

$$\frac{M(x)}{E(x)I(x)} = \frac{1}{\rho(x)} \tag{2}$$

According to the formula, when the structure is damaged, the flexural stiffness at the damaged part of the structure decreases, thus increasing the curvature at the damaged part, resulting in sudden change of the curvature mode shape. By comparing the curvature mode shapes before and after the damage, the damage condition of the structure can be identified. The curvature mode can be obtained by difference calculation on the basis of displacement mode measurement (Pandey et al., 1991):

$$\varphi_i''(x) = \frac{\varphi_i(j-1) - 2\varphi_i(j) + \varphi_i(j+1)}{h^2} \tag{3}$$

Where: $\varphi_i(j)$ is the i-th mode shape value of the j-th node; h is the distance between nodes.

Compared with the curvature mode, the curvature mode difference has a better ability to discriminate structural damage. Therefore, the curvature mode difference is chosen as the parameter index of damage identification in this paper. The absolute value expression of curvature mode difference is:

$$\Delta\varphi'' = |\varphi_d'' - \varphi_r''| \tag{4}$$

Where: $\Delta\varphi''$ is the curvature mode difference of the node;

φ_d'' is the curvature mode at the node of damaged structure;

φ_r'' is the curvature mode at the node of the undamaged structure.

In addition, the curvature mode ratio can also be used for damage

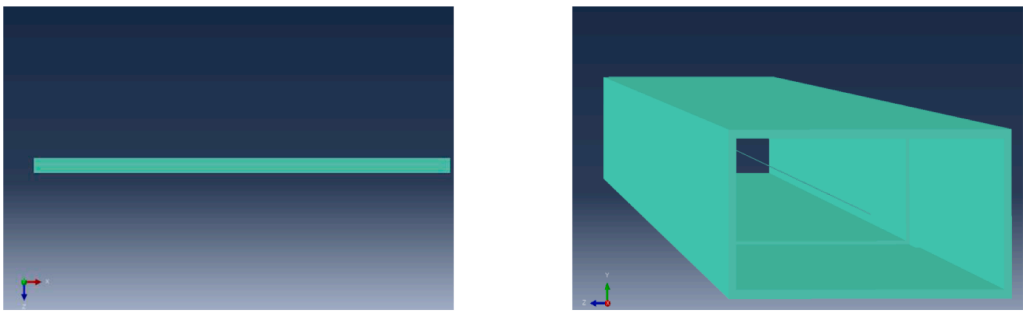


Fig. 3.1. Simulation model.

Table 3.1
Model damage condition number.

Single damage Degree of damage	Damage unit		Section plate thickness (mm)
Intact	Unit 9	Unit 15	5.5
15%	①	④	4.6
25%	②	⑤	4.0
45%	③	⑥	2.8
Double damage			
Degree of damage	Damage unit Units 5 and 9	Units 12 and 15	Section plate thickness (mm)
45%	⑦	⑧	2.8

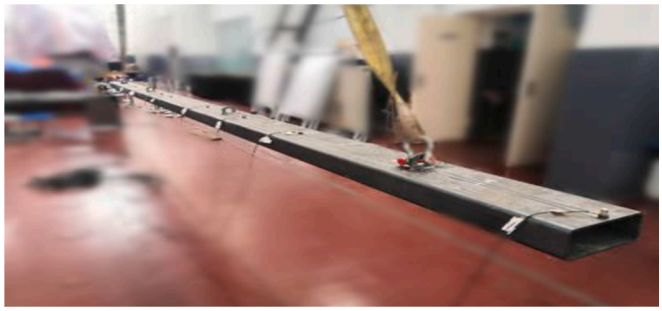


Fig. 3.2. Experimental model.

location identification of structures, expressed as:

$$R = \left| \frac{\varphi_d'' - \varphi_r''}{\varphi_d''} \right| \tag{5}$$

Where:R is the curvature mode ratio of the node;
φ_d'' is the curvature mode at the node of damaged structure;
φ_r'' is the curvature mode at the node of the undamaged structure.

The position corresponding to the larger curvature mode difference index value is the damage position of the hull beam. Flexible comparison of index changes at each location can realize damage identification more accurately. In practice, the peak value of curvature mode difference not only appears in the position near the damage, but also in the area far from the damage, which can lead to misjudgment. In order to eliminate the error of higher-order indicators, the average value of the first four orders of curvature mode difference index is given in this paper to realize the positioning of damage.

2.3. Damage identification theory based on modal flexibility

Modal flexibility is a fingerprint sensitive to structural damage (Li et al., 2012). The modal flexibility matrix is the inverse of the stiffness matrix, and only several low order modes are needed to obtain better accuracy. The calculation formula is as follows:

$$F = \Phi\Omega^{-1}\Phi^T = \sum_{i=1}^{n} \frac{1}{\omega_i^2}\phi_i\phi_i^T \tag{6}$$

Where:$\Phi = [\phi_1, \phi_2, \cdots, \phi_n]$ is mode matrix; ϕ_i is the i-th order mode;
$\Omega = diag(\omega_i^2)$ is a diagonal matrix; ω_i is the i-th frequency.

Structural damage results in decreased stiffness and increased flexibility, and the flexibility change matrix before and after damage is:

$$\Delta = F_i - F_d \tag{7}$$

Where: F_i is the flexibility matrix before structural damage;
F_d is the flexibility matrix after structural damage.

The modal flexibility change rate matrix is obtained by comparing the modal flexibility change matrix with the flexibility matrix before structural damage:

$$RC = \frac{\Delta}{F_i} \tag{8}$$

The difference method was used to calculate the curvature of the modal flexibility change rate matrix, and the curvature index of the modal flexibility change rate was obtained:

$$MFCRC = \frac{RC(j-1) - 2RC(j) + RC(j+1)}{h^2} \tag{9}$$

Table 3.2
Model damage condition number.

Damage unit Damage degree	Unit 9	Unit 15
15%	①	④
25%	②	⑤
45%	③	⑥

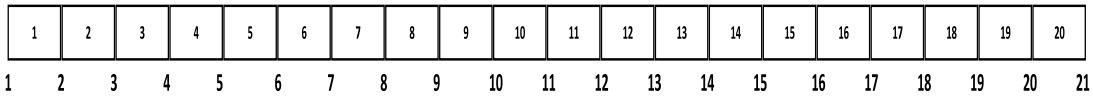


Fig. 3.3. Node numbers.

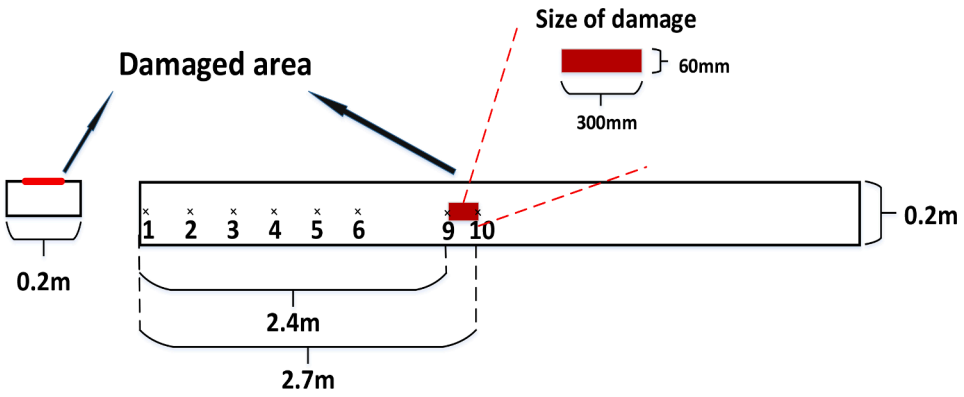


Fig. 3.4. Schematic diagram of stiffness loss simulation.

Table 3.3
Fracture size and stiffness loss.

Damage simulation	Fracture 1	Fracture 2	Fracture 3
Sketch map			
Fracture size	300mmx60mm	300mmx96mm	300mmx146mm
Stiffness loss	15%	25%	45%

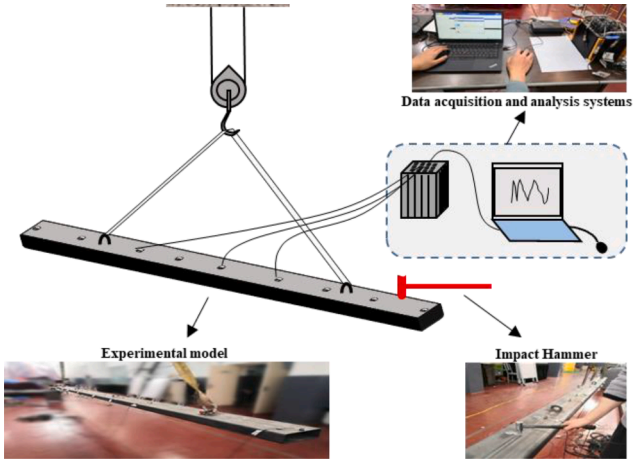


Fig. 3.5. Schematic diagram of test design.

Where: j is node number; h is the distance between nodes.

The position corresponding to the larger value of the curvature index of the modal flexibility change rate is the damage position of the hull beam. Flexible comparison of index changes at each location can realize damage identification more accurately.

2.4. Damage degree identification theory based on perturbation method

Based on Euler-Bernoulli equation, the vibration differential equation of hull beam is given:

$$\frac{\partial^2}{\partial x^2}\left\{EI(x)\frac{\partial^2[v(x,t)]}{\partial x^2}\right\}+\rho(x)\frac{\partial^2}{\partial t^2}[v(x,t)]=0 \tag{10}$$

Where: $EI(x)$ is the bending stiffness of hull beam;
$v(x,t)$ is the longitudinal displacement of hull beam;
$\rho(x)$ is the density of hull beam per unit length.

In this paper, only the local stiffness loss is considered to simulate the structural damage of hull beam, and the mass loss is ignored. The separation variable method is adopted to solve the vibration differential equation:

$$\frac{\partial^2}{\partial x^2}\left\{EI_d(x)\frac{\partial^2[\phi_i^d(x)]}{\partial x^2}\right\}-\rho(x)\lambda_i^d[\phi_i^d(x)]=0 \tag{11}$$

Where: λ_i^d Is the characteristic value of the equation solution of the damaged beam; $\phi_i^d(x)$ is the characteristic vector of the solution of the damaged beam equation, namely the mode function; $EI_d(x)$ is the damage stiffness of hull beam, and the relationship between the stiffness of intact hull beam is as follows:

$$EI_d(x)=EI_i(x)\{1-\varepsilon[H(x-x_1)-H(x-x_2)]\} \tag{12}$$

Where: $EI_i(x)$ is the intact stiffness of hull beam;
ε is the degree of damage$(0<\varepsilon<1)$ for a small amount;
$H(x-x_i)$ is the unit step function;
x_1 and x_2 are the starting point and end point of the damaged area.

For the smaller ε, it is expanded by perturbation method, and the asymptotic solution is as follows (Zhang et al., 2022):

$$\begin{aligned}\lambda_i^d&=\lambda_i^0-\varepsilon\lambda_i^1+o(\varepsilon^2)\\ \phi_i^d(x)&=\phi_i^0(x)-\varepsilon\phi_i^1(x)+o(\varepsilon^2)\end{aligned} \tag{13}$$

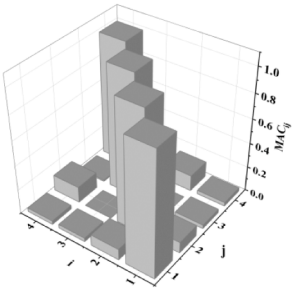
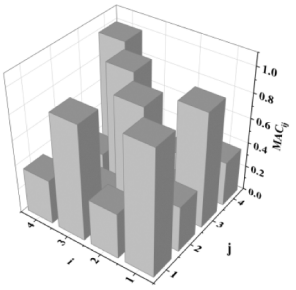
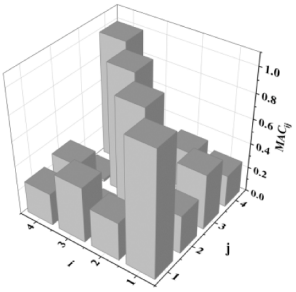
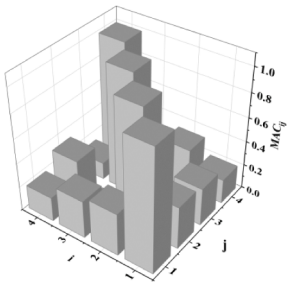
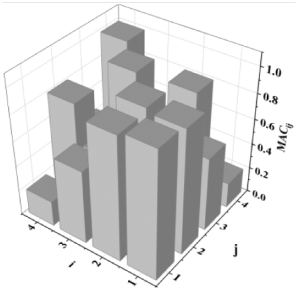
The governing equation of a damaged hull beam is obtained by substituting it into the differential equation of a vibrating hull beam:

$$\varepsilon^0: EI_i\frac{\partial^4}{\partial x^4}[\phi_i^0(x)]-\lambda_i^0\rho\phi_i^0(x)=0 \tag{14}$$

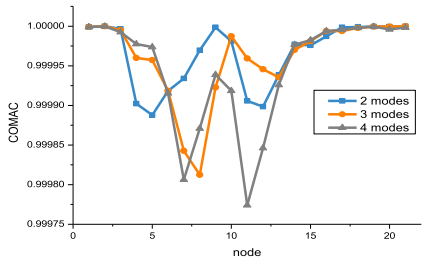
$$\begin{aligned}\varepsilon^1: &EI_i\frac{\partial^4}{\partial x^4}[\phi_i^1(x)]-\lambda_i^0\rho\phi_i^1(x)\\ &=-EI_i\frac{\partial^2}{\partial x^2}\left\{[H(x-x_1)-H(x-x_2)]\frac{\partial^2}{\partial x^2}[\phi_i^0(x)]\right\}+\lambda_i^1\rho\phi_i^0(x)\end{aligned} \tag{15}$$

The modal shape of the damaged hull beam structure is:

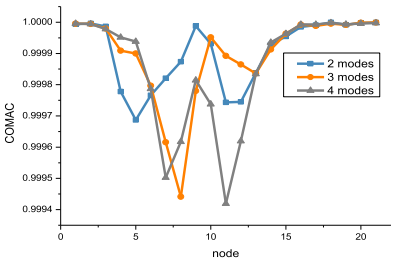
Table 3.4
Model test scheme index values.

scheme	MAC diagram	measure point	evaluation index value
optimal		15,791,213,151,721	4.58
verification 1		24,691,112,141,620	8.7
verification 2		12,491,112,141,621	6.76
verification 3		1467 8,913,141,721	6.66
verification 4		35,691,011,151,820	11

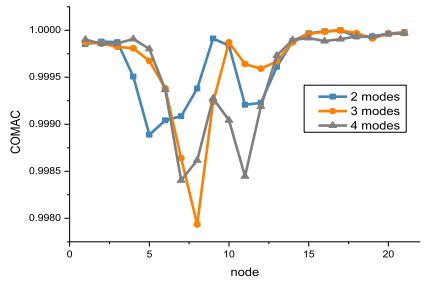
$$\begin{aligned}\phi_i^d(x) &= \phi_i^0(x) - \varepsilon\phi_i^1(x)\\ &= \phi_i^0(x) - \varepsilon\{\sum_{j=1,j\neq i}^{n}\frac{\lambda_i^0}{\lambda_i^0-\lambda_j^0}\left(G_{1j}-G_{2j}+G_{3j}-G_{4j}-G_{5j}\right)\phi_j^0(x) + < x-x_1 > \frac{\partial}{\partial x}\phi_i^0(x)|_{x=x_1}\\ &- < x-x_2 > \frac{\partial}{\partial x}\phi_i^0(x)|_{x=x_2} - [H(x-x_1)-H(x-x_2)]\phi_i^0(x)\}\end{aligned} \tag{16}$$



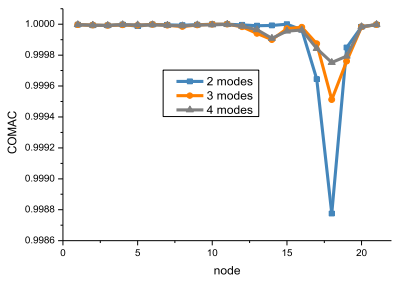
(a) Damage 1



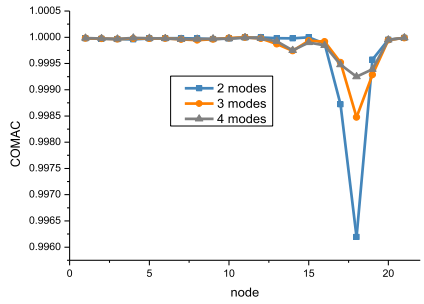
(b) Damage 2



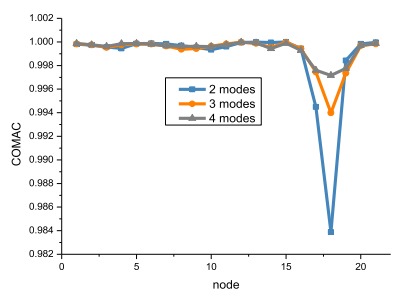
(c) Damage 3



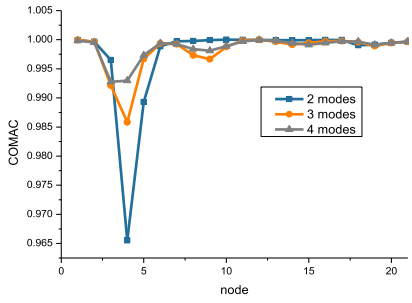
(d) Damage 4



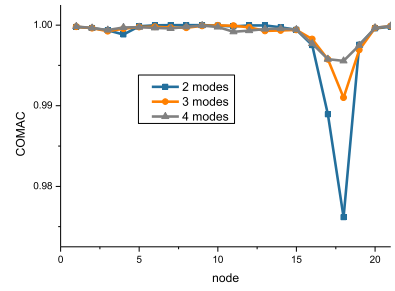
(e) Damage 5



(f) Damage 6

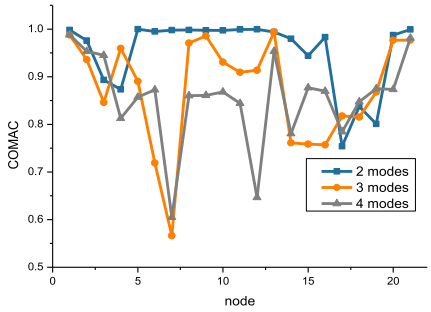


(g) Damage 7

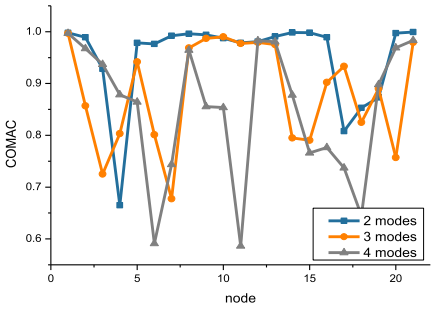


(h) Damage 8

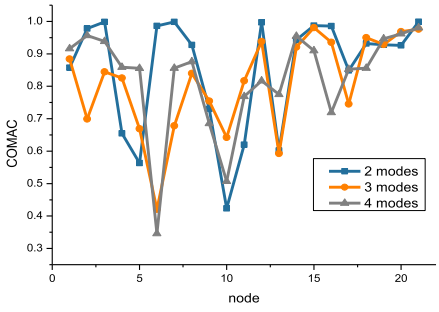
Fig. 4.1. Simulation of COMAC index for each working condition.



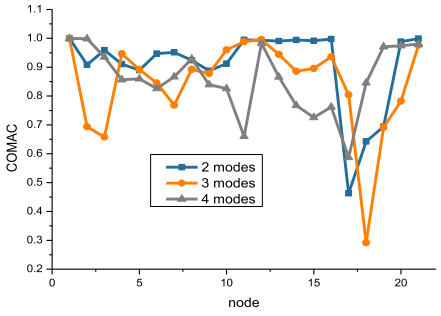
(a) Damage 1



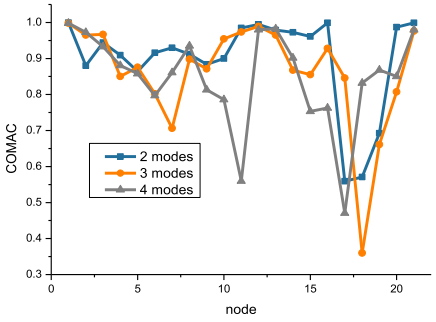
(b) Damage 2



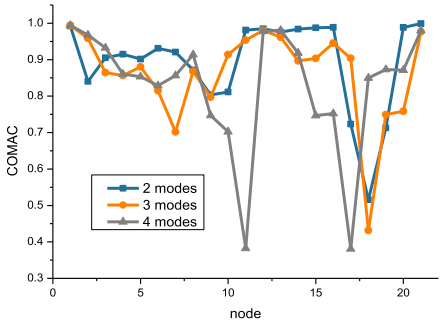
(c) Damage 3



(d) Damage 4



(e) Damage 5



(f) Damage 6

Fig. 4.2. Experiment of COMAC index for each working condition.

When the hull beam structure is damaged, the curvature mode difference is obtained, and then the damage degree is obtained:

$$\varepsilon = -\Delta \Bigg/ \left\{ \sum_{j=1,j\neq i}^{n} \frac{\lambda_i^0}{\lambda_i^0 - \lambda_j^0} \left(G_{1j} - G_{2j} + G_{3j} - G_{4j} - G_{5j}\right) \left[\phi_j^0(x)\right]'' - \left[H(x-x_1) - H(x-x_2)\right] \left[\phi_i^0(x)\right]'' \right\} \tag{17}$$

Where: Δ is the modal difference of hull beam curvature;

$$G_{1j} = \frac{\partial}{\partial x}\phi_i^0(x)|_{x=x_1} \int_0^L < x - x_1 > \phi_j^0(x) d(x);$$

$$G_{2j} = \frac{\partial}{\partial x}\phi_i^0(x)|_{x=x_2} \int_0^L < x - x_2 > \phi_j^0(x) d(x);$$

$$G_{3j} = \phi_i^0(x_1) \int_0^L H(x - x_1)\phi_j^0(x) d(x);$$

$$G_{4j} = \phi_i^0(x_2) \int_0^L H(x - x_2)\phi_j^0(x) d(x).$$

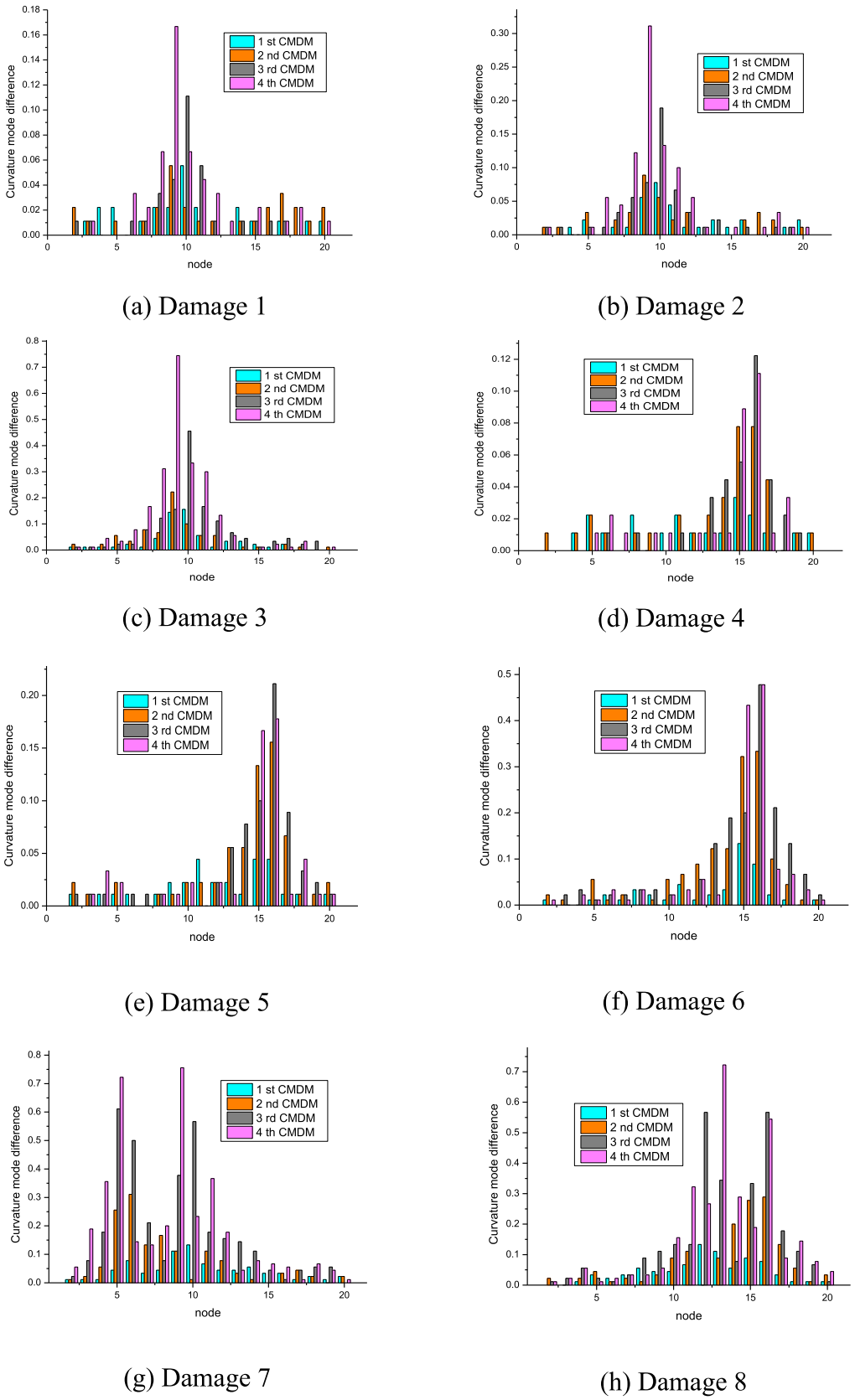


Fig. 4.3. Simulation of the mode difference of each order of curvature for each working condition.

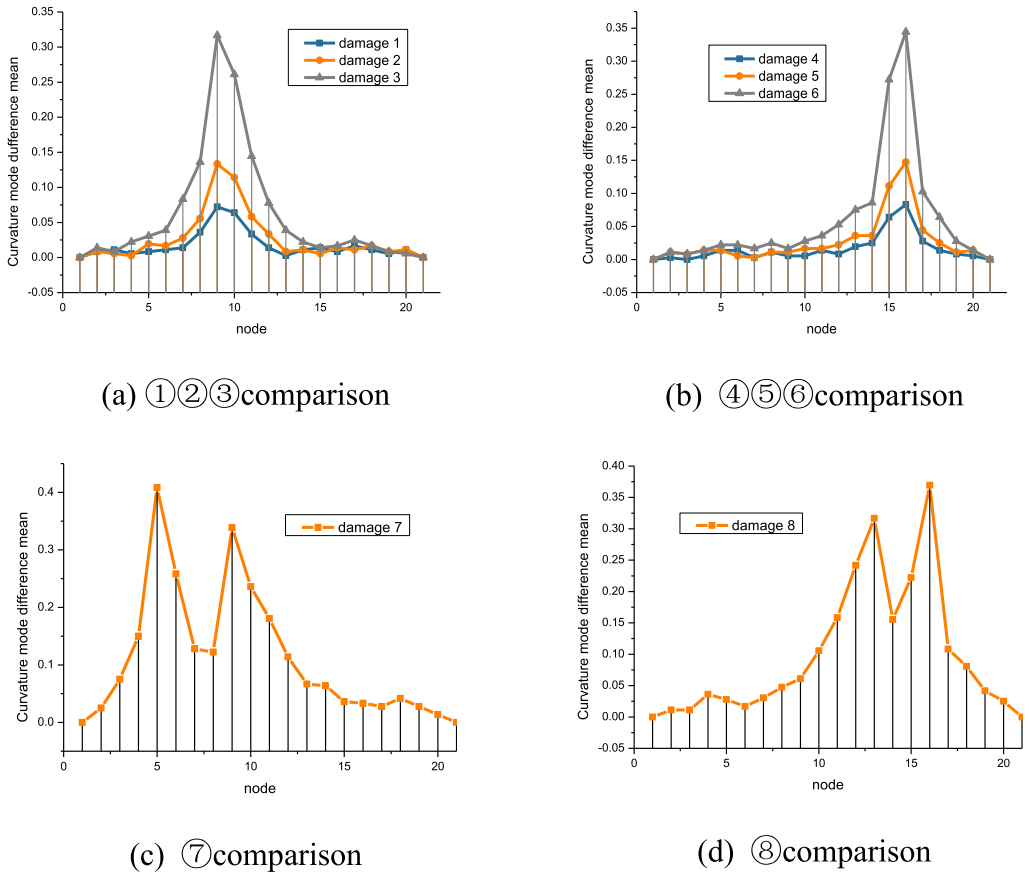


Fig. 4.4. Simulation of the first four orders of curvature mode difference mean value for each working condition.

3. Simulation and experimental models

3.1. Finite element simulation model

The numerical simulation model in this paper is 6 m in length, 0.2 m in width and 0.1 m in height. With line modeling attached to the section properties, a total of 20 segments and 21 nodes are included, each unit length is 0.3 m, the section shape is rectangular, and the plate thickness is 5.5 mm, as shown in Fig. 3.1.The modulus of elasticity is 2.06E+11pa, density is 7850 kg/m3, Poisson's ratio is 0.3, and the boundary conditions are fully free boundary conditions. According to the vibration equation of hull beam structure,$(K - \lambda_i M)\phi_i = 0$, the hull beam vibration pattern is determined by the modal displacements of the vertical degrees of freedom at 21 nodes.

Three damage conditions are set up in unit 9 and unit 15 of the model, all of which are single damage. In order to verify the effectiveness of each method for multi-damage identification, two additional double-damage conditions are used for finite element simulation analysis, as shown in Table 3.1. Different cross-sectional dimensions of the corresponding units are adopted to reduce the cross-sectional moment of inertia, simulate the stiffness loss of the hull structure, and perform modal analysis on it. Reads the frequencies and vibration patterns of the simulation model for the intact and eight operating conditions.

3.2. Experimental model

The dimensions of the experimental model in this paper are the same as the simulation, as shown in Fig. 3.2, with a total weight of 169.96 kg and a material density of 7850 kg/m3, containing a total of 20 segments and 21 nodes, as shown in Fig. 3.3. In this paper, six beam models with different damage conditions are designed, namely, three damage conditions are set respectively in unit 9 and unit 15 of the model, as shown in Table 3.2. Notches of different sizes are set for corresponding units to simulate stiffness loss, as shown in Fig. 3.4, Table 3.3.

In the simulation of the fully free boundary conditions, the suspension support method is used to suspend the structure freely in the fixed space with elastic ropes, and the boundary conditions formed in this way cause little effect on the structure. In addition, in order to further improve the effectiveness of the boundary condition simulation, the position of the lifting point is analyzed according to the modal characteristics of the test model, and the lifting point is positioned at the position of high-order modal nodes, so as to reduce the influence of the test conditions.

The Donghua5902 modal test system, consisting of a data acquisition instrument, a laptop with built-in modal analysis software and a force hammer, is used in the experiment in this paper.

According to the modal characteristics of the model, nine acceleration sensors are used to conduct modal tests on the model, as shown in Fig. 3.5.

3.3. Points optimization

For a large structure, damage identification requires a large number of sensors, and the large number of sensors brings maintenance burden, increased expenses and massive data, so the optimization of sensor measurement points is an important issue. Considering the beam structure characteristics of hull beams, this paper selects the modal assurance criterion (MAC) as the evaluation index of the characteristic signal optimization test scheme, and on this basis carries out the research on the optimization of the test scheme of hull beams under the optimization configuration criterion. MAC is calculated as follows:

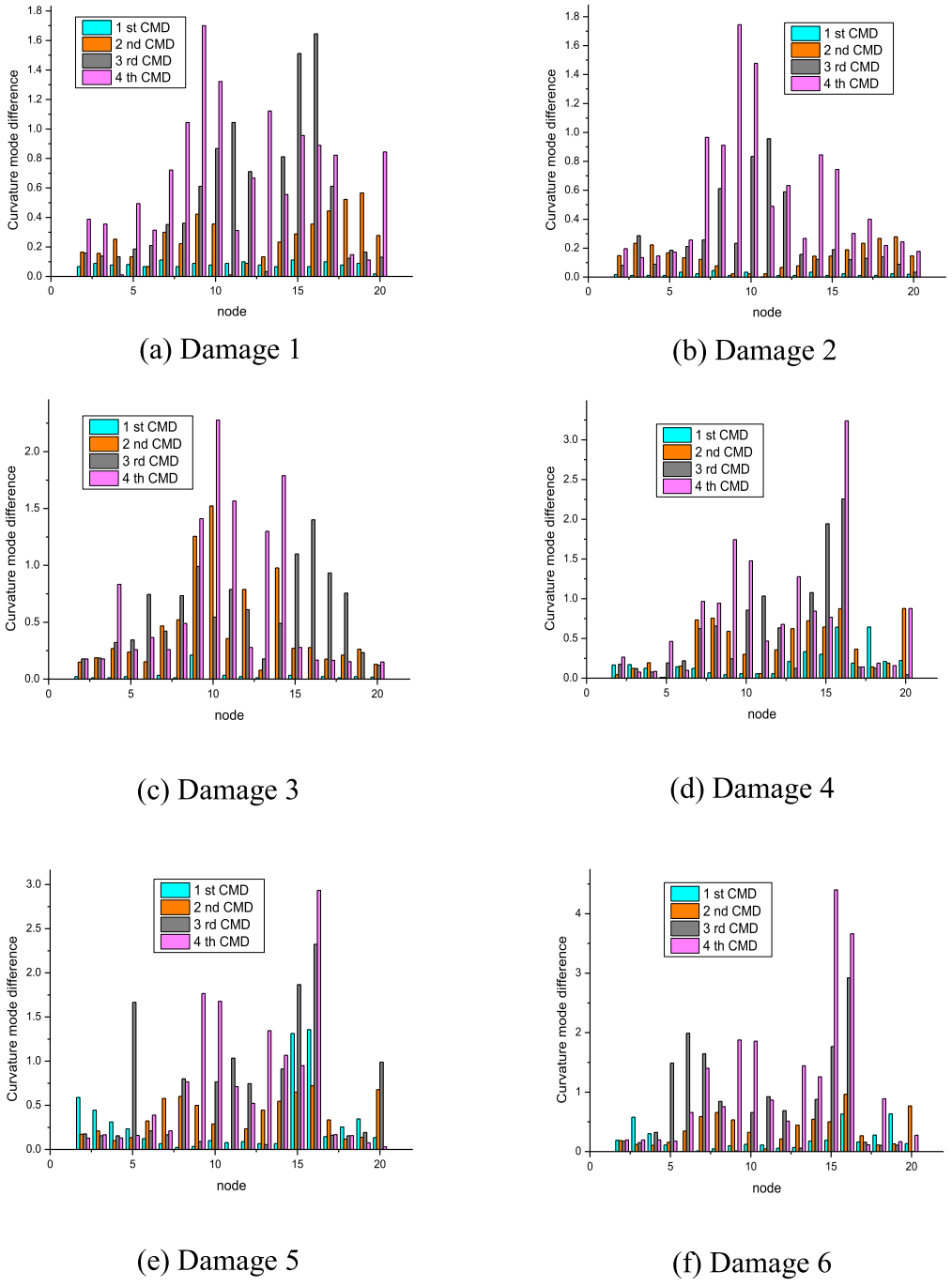


Fig. 4.5. Experiment of the mode difference of each order of curvature for each working condition.

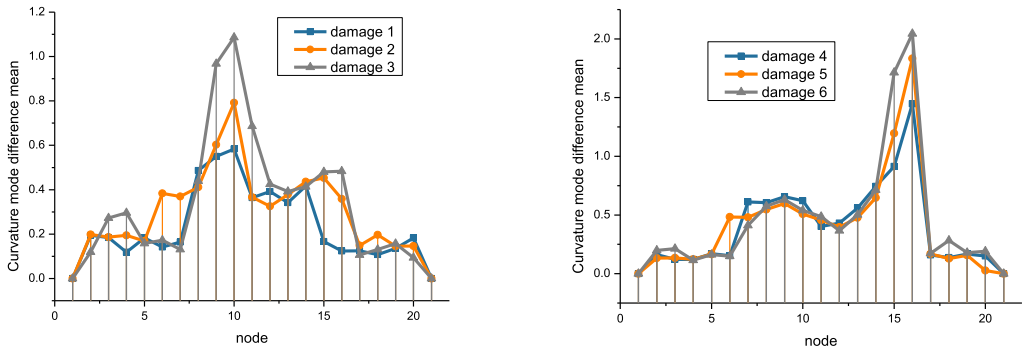
$$MAC_{ij} = \frac{\left[\left(\varphi_i^u\right)^T \varphi_j^d\right]^2}{\left[\left(\varphi_i^u\right)^T \varphi_i^u\right]\left[\left(\varphi_j^u\right)^T \varphi_j^u\right]} \tag{18}$$

Where, $\{\phi_i\}$and$\{\phi_j\}$ represent mode vectors of order i and order j respectively.

In this paper, the sum of all the elements of the MAC matrix is used as the evaluation index for the optimization of the feature signal test, and the measurement point placement scheme that minimizes the sum of all the elements of the MAC is obtained by iteration.

The basic principle of measuring point layout is to avoid the location of mode nodes and to locate at the amplitude point of mode as far as possible. The initial calculation of the vibration parameters of the model according to the finite element simulation, combined with the vibration amplitude points and node positions, the initial measurement point scheme obtained is 1, 5, 7, 9, 13, 15, 17, 18, 21. Considering the actual engineering application of the site measurement point deployment implementation conditions, some measurement point locations can not be sensor installation, so this paper will consider the impact of measurement point offset on the test results. The available measurement point options are 1 (2), (4) 5 (6), 7 (8), 9 (10), (12) 13 (14), 15 (16), 17, 18 (19), (20) 21, with alternative measurement points in parentheses. After the arrangement and combination, 576 test measurement point solutions can be obtained.

Next, the above measurement point schemes are combined with the evaluation indexes to be preferred. In this paper, the algorithm is written according to the above requirements and the optimal measurement point scheme is obtained as shown in Table 3.4. In order to verify the



(a) ①②③comparison (b) ④⑤⑥comparison

Fig. 4.6. The mean value of the difference in curvature mode of the first four orders for each working condition of the experiment.

validity of the scheme, four additional sets of validation measurement point schemes are set up to identify and analyze the modal information of the test model using a single excitation method. In order to prevent a certain degree of error in the calculation of the frequency response function during a single test, this test uses five excitations of the test model to collect the dynamic response signal to calculate its frequency response information and perform linear averaging.

As can be seen from the table, the evaluation index value of the best test scheme is the smallest, which reflects that the vibration patterns obtained from the test under this test scheme have good orthogonality and meet the theoretical requirements, thus verifying the effectiveness of the evaluation method of the test scheme in this paper.

4. Analysis of results and discussion

The modal analysis is performed on the finite element simulation model and the experimental model in Section 3 to obtain the frequencies and vibration patterns, which are brought into the self-programmed algorithms based on COMAC, curvature mode difference, curvature mode ratio and modal flexibility, respectively, for damage identification study.

4.1. Damage identification using COMAC

According to the theory, the location of the damage to the hull structure should correspond to the minimum value at the COMAC indicator. The results of the finite element and experimental damage identification using COMAC are shown in Figs. 4.1 and 4.2.

The results of the simulation analysis in Fig. 4.1 show that: The minimum COMAC values under damage 1, 2 and 3 conditions are on both sides of node 9, the first three and four orders of modal COMAC indicators are within two units of the actual damage unit 9, and the first two orders of modal COMAC indicators have large errors; The COMAC minimums for damage 4, 5 and 6 conditions are at nodes 17 and 18, which differ by two units from the actual damage unit 15; For the double damage 7 and 8 conditions, the index does not identify either unit of the damage.

The results of the experimental validation in Fig. 4.2 show that: The minimum COMAC values for damage 1, 2, and 3 conditions are between node 5 and node 10, and the minimum COMAC values for damage 4, 5, and 6 conditions are between node 15 and node 18. The first two, three, and four order modal COMAC cannot be accurately located to the damage unit.

From a combination of simulation and test results, it is clear that the COMAC index cannot accurately identify the specific unit where the damage existsIn terms of damage degree, the COMAC index gradually decreases as the damage degree increases, but the degree of differentiation is not high and it does not accurately reflect the true damage degree of the structure.

4.2. Damage identification using curvature mode difference

According to the theory, the maximum value of the curvature mode difference of each order should correspond to the damage location of the hull structure. The results of finite element and experimental damage identification using curvature mode differences are shown below.

The results of the simulation analysis in Fig. 4.3 show that: The difference in curvature modes of each order under working conditions 1, 2 and 3 is greatest near unit 9, and it can be clearly seen that the difference in curvature modes of order 2 and 4 precisely achieves the localization of damage, and the indexes of order 1 and 3 are localized in unit 10, which is close to the actual damage unit; Differences in curvature modes of all orders under working conditions 4, 5 and 6 are precisely localized for damage, in unit 15; The location of the double damage is accurately identified for the 3rd and 4th order curvature mode difference in both working condition 7 and working condition 8, with the 5th and 9th damage units identified in working condition 7 and the 12th and 15th damage units identified in working condition 8.

The curvature mode differences for the first four orders are taken as the mean value, and the comparison results are shown in Fig. 4.4.

The results of the simulation analysis in Fig. 4.4 show that: For each working condition, the first four orders of curvature modal difference at the peak value of the mean corresponds to the hull structure damage location, and the identification of the damage unit is accurate. As the degree of unit damage increases, the index gradually increases at the location of the damage, and the degree of differentiation is obvious.

The results of the experimental validation in Fig. 4.5 show that: Under working conditions 1 and 2, it is obvious that the 4th order curvature mode difference accurately achieves the damage localization; Under working conditions 3, the 2nd order curvature mode difference is located close to the actual damage unit; The curvature mode difference of each order under the working conditions 4, 5 and 6 accurately realizes the damage location, which is in unit 15. In summary, the experiment and simulation are the same, the fourth order curvature mode difference identification is the most accurate, compared with other third orders, can more clearly see the damage location.

The mean values of the first four order curvature mode differences of different damage degrees of the same damage element in the test are compared, and the results are shown in Fig. 4.6.

The results of the experimental validation in Fig. 4.6 show that: Under 6 working conditions, the peak value of the mean difference of the first four curvature modes corresponds to the damage unit, and the identification results are accurate. The higher the degree of damage, the more obvious the recognition result. With the increase of the damage

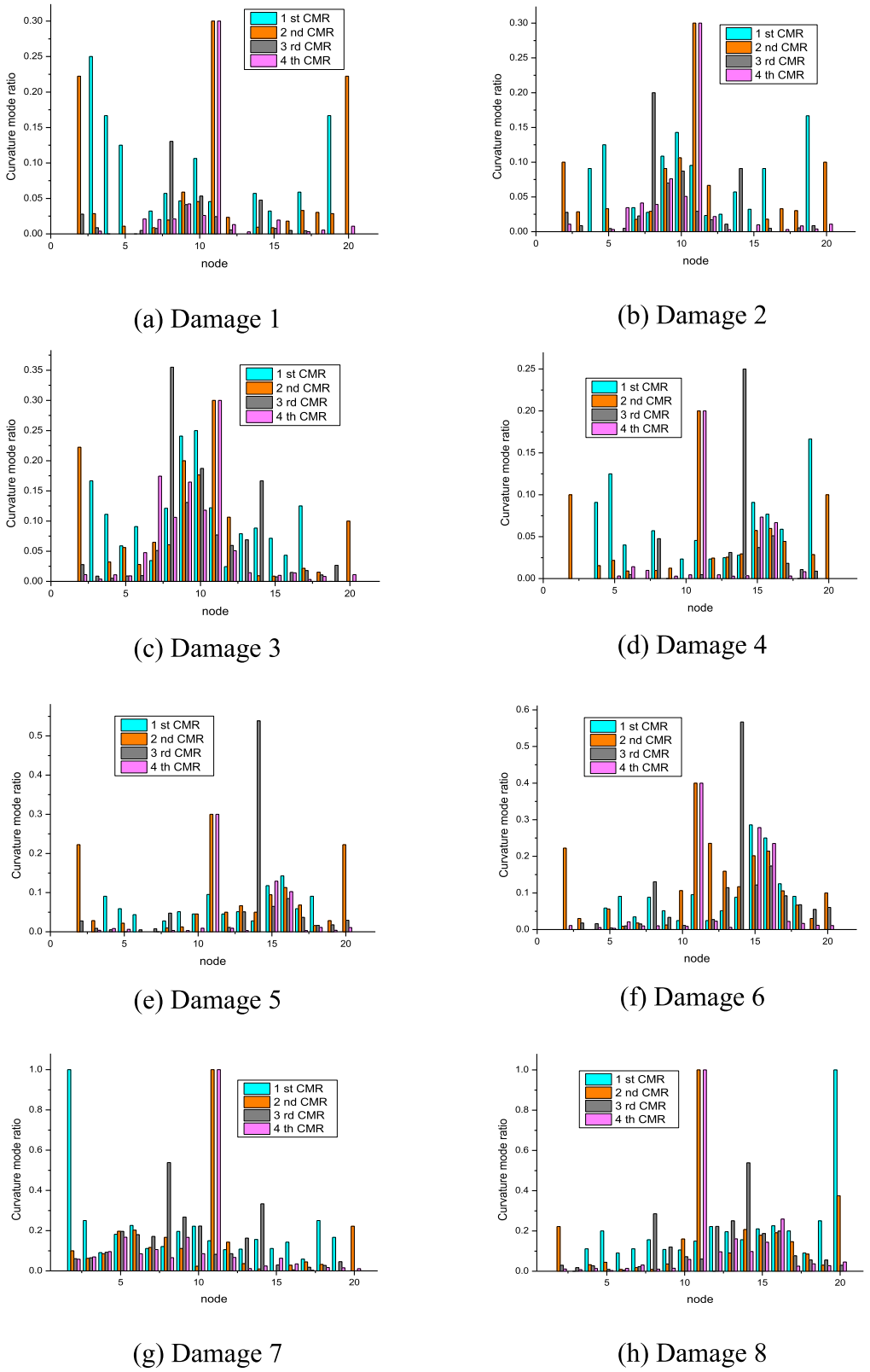


Fig. 4.7. Simulation of each order of curvature mode ratio index for each condition.

degree of the unit, the mean curvature mode difference increases gradually at the damage position, and the difference is obvious.

From the simulation and test results, it can be seen that the curvature mode difference index can accurately identify the specific unit where the damage exists, and the mean value index of the first four order curvature mode difference has a positive correlation with the damage degree. Therefore, the degree of damage can be identified by the degree of change of this index. Compared with the simulation, the mean value of the first four order curvature mode difference corresponding to the test value not only has a peak value at the damage location, but also near and

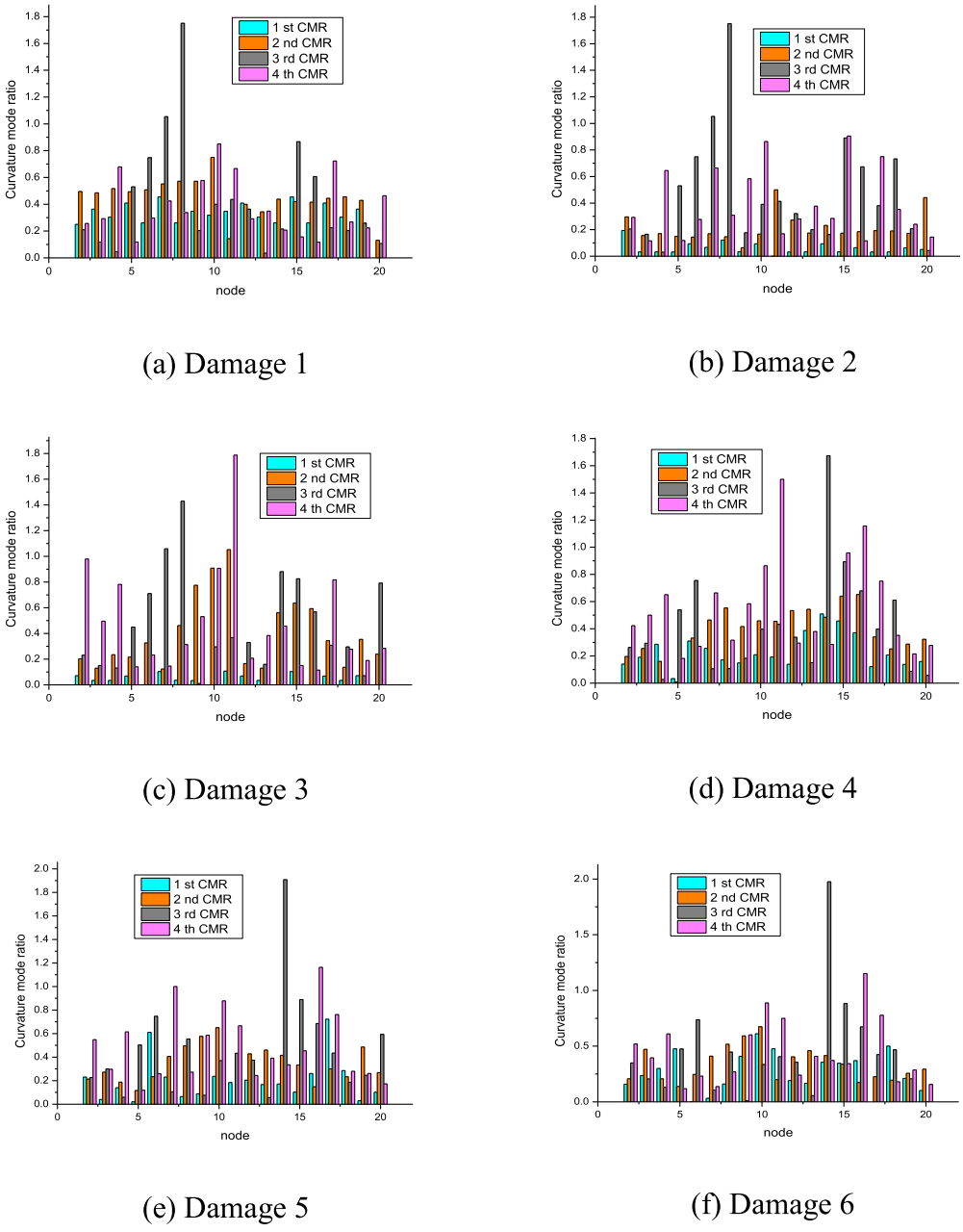


Fig. 4.8. Experiment of each order of curvature mode ratio index for each condition.

far from the damage area, which is considered to be related to the vibration mode change caused by interference noise and suspension position.

4.3. Damage identification using curvature mode ratio

According to the theory, the maximum value of the curvature mode ratio at each order should correspond to the damage location of the hull structure. The results of the finite element and experimental damage identification using this index are shown in Figs. 4.7 and 4.8.

The results of the simulation analysis in Fig. 4.7 show that: For single damage condition, only when the damage degree of the first order curvature mode ratio is 45%, the maximum value corresponds to the actual damage unit, while the other orders curvature mode ratio can not identify the actual damage unit under any condition. For the double damage condition, any order of curvature mode ratio cannot accurately identify the damage unit.

The results of the experimental validation in Fig. 4.8 show that: Only the maximum value of the third order curvature mode ratio is near the actual damage unit, but there are more interference peaks at the position far away from the damage unit, so it is impossible to accurately identify single damage. The curvature mode ratios of order 1, 2 and 4 cannot identify the damage unit.

Comprehensive simulation analysis and test verification show that there are individual indexes corresponding to the actual damage location when identifying single damage for each curvature mode ratio, but they are accidental and have no obvious rules. Moreover, the double damage could not be identified for each curvature mode ratio. Therefore, the curvature mode ratio index can not accurately identify the specific unit where the damage exists.

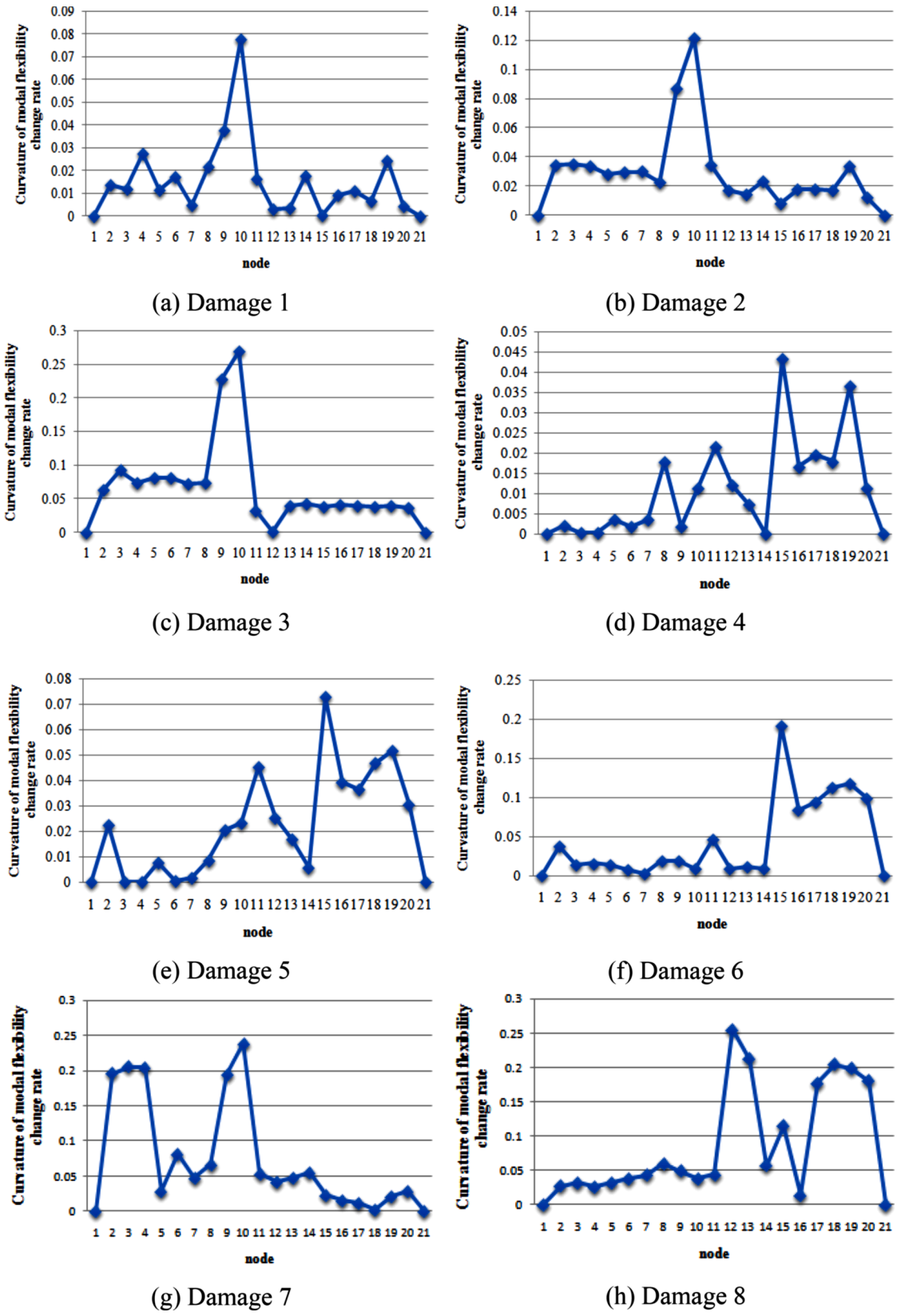


Fig. 4.9. Curvature index of modal flexibility change rate for each condition of the simulation.

4.4. Damage identification using the curvature of modal flexibility change rate

According to the theory, the maximum curvature of the modal flexibility change rate should correspond to the damage position of the hull structure. The results of finite element and test using the curvature of modal flexibility change rate for damage identification are shown in Figs. 4.9 and 4.10.

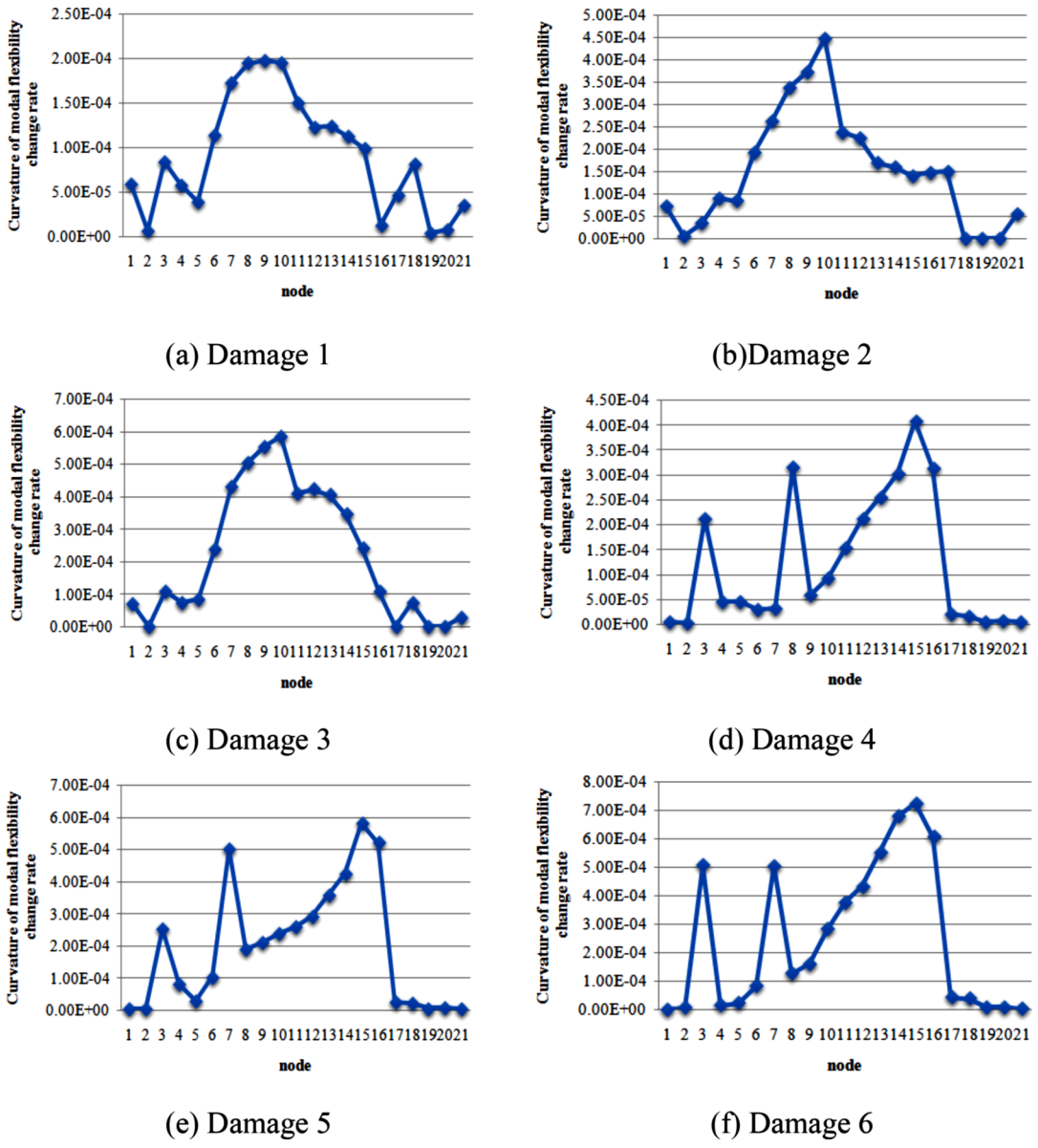


Fig. 4.10. Curvature index of modal flexibility change rate for each condition of the experiment.

Table 4.1
Results of damage degree identification.

condition	Damage unit	Actual damage degree	Simulation identification		Experiment identification	
			Degree	Error	Degree	Error
①	Unit 9	15%	18.53%	23.53%	25.56%	70.40%
②		25%	26.30%	5.20%	39.69%	58.76%
③		45%	47.34%	5.20%	73.36%	63.02%
④	Unit 15	15%	23.17%	54.47%	17.33%	15.53%
⑤		25%	41.70%	66.80%	38.79%	55.16%
⑥		45%	83.40%	85.30%	86.34%	91.87%

The results of the simulation analysis in Fig. 4.9 show that: In the condition of damage 1, 2 and 3, the damage location is identified in unit 9, and there is not too much interference peak. In the condition of damage 4, 5 and 6, the damage location is identified at unit 15, but there are more peak interference near and far from the damage unit. In the condition of double damage 7 and 8, only one damage unit can be accurately identified, and the other location is close to the damage unit. From the peak point of curvature index of modal flexibility change rate, it can be seen that with the increase of damage degree of the same damage unit, the value of this index gradually increases at the damage position, and the difference is obvious.

The results of the experimental validation in Fig. 4.10 show that: Under the 6 working conditions, the maximum curvature of the modal flexibility change rate corresponds to the actual damage unit, but there is an obvious interference peak, and it is impossible to determine whether there is only one damage. Compared with the peak point of the index of different damage degrees of the same damage unit, the index increases with the increase of damage degrees, but the magnitude is smaller than the curvature mode difference.

Comprehensive simulation analysis and test verification show that although the curvature index of modal flexibility change rate can correspond to the actual damage unit, there are many obvious interference peaks. If the damage identification is carried out only according to this index, it may lead to misjudgment.

4.5. Damage level identification using curvature mode difference

The first four orders of curvature mode difference obtained in simulation and experiment are respectively brought into the self-designed algorithm for damage degree identification research. The results are shown in Table 4.1:

Table 4.1 shows: For the simulation results, the damage degree of unit 9 is more accurate than that of unit 15, but it is opposite for the experiment results. The simulation identification results are the same as the experiment identification results, the error of the condition identification results of unit 9 with smaller damage degree is larger, and the error of the condition identification results of larger damage degree is smaller. The error of unit 15 damage identification results increases with the actual damage level. Whether in simulation or experiment, the degree of recognition differs significantly from the actual degree of damage, and overall, the error in the experimental recognition is somewhat greater. The reason for the above phenomenon may be that there are some errors in the mode parameters obtained by the suspension method to simulate the free boundary condition test, and the number of measurement points is too small, so there are some differences between the fitting mode and the actual model mode.

5. Conclusion

No matter for military ship or civilian ship, the change of hull structural damage will bring adverse effects, and the specific damage position and certain degree may directly lead to the sinking of the ship. An indirect hull damage identification technology and method is urgently needed in engineering to effectively identify ship structural damage. In this paper, based on the modal shape of the structure, COMAC method, the derivative index of curvature mode and modal flexibility are used to identify the damage of hull beam. The main conclusions of this research are as follows:

(1) When the curvature mode ratio index is used to identify single damage, there are cases where individual order locating damage is accurate, but it has randomness and instability, and double damage cannot be accurately identified. This index is not suitable for damage identification of hull beams.
(2) The curvature mode difference index needs to be used in combination with the data of each order to identify the damage location of hull beam structure more accurately. However, in terms of the degree of damage identification, the degree of identification is quite different from the actual degree of damage. In the next step, it is necessary to increase the number of measuring points appropriately, optimize the location of measuring points, and obtain more accurate vibration mode data of the test model.
(3) Although the peak value of the curvature index of modal flexibility change rate can correspond to the actual damage unit, there are many interference peaks, it is impossible to determine whether it is a single damage or multiple damage, and only one can be accurately identified when identifying two damages. The index showed a positive correlation with the degree of damage. In the next step, this index can be optimized to reduce the interference peak, and further study the use of this index for damage degree identification.

Author contribution statement

Haoyue Cheng: Development or design of methodology; creation of models; Preparation, creation and/or presentation of the published work, specifically writing the initial draft (including substantive translation).

Jun Guo: Oversight and leadership responsibility for the research activity planning and execution, including mentorship external to the core team.

Yin Zhang: Preparation, creation and/or presentation of the published work, specifically visualization/ data presentation.

Wenzhuo Li: Reproducibility of results and other research outputs.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled, “Experimental research on damage identification of fully free beams based on modal shapes”.

Data availability

Data will be made available on request.

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